

LLM-Based Multi-Agent Orchestration: A Survey of Frameworks, Communication Protocols, and Emerging Patterns

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

3-topology / 1-adaptivity coordination taxonomy comparing state management, token expenditure, and failure recovery.

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Literature corpus 2023–early 2026 across 6 frameworks.

Key Numerical Findings & Empirical Proofs:

Proves token consumption scales as $O(M * R)$; establishes decision criteria for selecting coordination topologies.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Assumes problem difficulty is known in advance; does not provide an autonomous runtime trigger for switching between solo and debate graphs.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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